

Anti-Dilution Tool Calibration Methodology

How the Firm calculates and validates swing factors, dilution levies and dual-pricing adjustments, asset class by asset class, position by position

Firm	Anchorpoint Fund Managers (UK) Limited — £210bn UK-authorised range, 62 funds
Model name / ID	Anti-Dilution Cost Estimation Model ("the Model"), Model Register ID LIQ-ADT-001
Model owner (first line)	Head of Liquidity Risk
Document owner	Head of Liquidity Risk, approved by the Chief Risk Officer
Independent validator (second line)	Model Validation function, independent of Risk and Investment reporting lines
Model risk tier	Tier 2 — Price-Adjacent Model (Section 3)
Approving body	Liquidity Risk Committee, following Model Validation sign-off
Version	5.0 — PS26/17 implementation (supersedes v4.2)
Effective date	1 December 2026
Next full revalidation due	December 2027 (annual), or immediately on a Red backtesting outcome (Section 13)
Parent document	Fund Liquidity Risk Policy, Section 6.3 · Model Risk Policy
Classification	Internal — Risk, Investment and Model Validation; parameters restricted to the Liquidity Risk Committee

1. Purpose and scope of this document

1.1 This document sets out the method by which Anchorpoint Fund Managers (UK) Limited — manager of a £210 billion UK-authorised fund range spanning equity, fixed income, multi-asset, money market and real asset strategies — calculates the anti-dilution mechanism (a swing factor, dilution levy, or dual-pricing dealing charge) required by COLL 6.3.7AR, calibrated in accordance with COLL 6.3.7CG and the IOSCO guidance on anti-dilution liquidity management tools. It is submitted as the model methodology document for LIQ-ADT-001 under the Firm's Model Risk Policy and is the primary artefact against which Model Validation forms its opinion.

1.2 This document covers: the conceptual basis of the Model (Section 4); its functional form (Section 5); the calibration basis (Section 6); a fully worked numerical example (Section 7); scaling and asymmetry behaviour (Section 8); the stress overlay (Section 9); asset-class parameterisation (Section 10); data lineage (Section 11); the parameter estimation methodology (Section 12); the backtesting framework (Section 13); sensitivity analysis (Section 14); known limitations (Section 15); ongoing monitoring and change control (Section 16); and the model risk rating and validation record (Section 17).

1.3 This document does not cover governance of the decision to activate the mechanism on a given day (Fund Liquidity Risk Procedure, Section 5) or the maker-checker control over its application (same Procedure, Section 5.1). It is exclusively concerned with how the rate itself is derived.

2. Model overview

2.1 The Model estimates the cost a fund would incur, at a given valuation point, in converting scheme property to cash (for a redemption) or in deploying new cash into scheme property (for a subscription), sufficient to allow that cost to be charged to the transacting unitholder rather than borne by unitholders who remain invested. It produces a single output — a rate, expressed in basis points of unit price — for a given fund and a given net flow.

2.2 The Model is a first-line tool operated by the Liquidity Risk function. It is not a valuation model: it does not determine the fund's net asset value. It is, however, "price-adjacent" in that its output, once signed off under the maker-checker control, directly adjusts the price at which units are issued or cancelled, which is why it is model-risk-managed to the standard set out in this document rather than treated as an unvalidated risk report.

3. Model risk tiering

3.1 The Model is classified **Tier 2 — Price-Adjacent Model** under the Firm's Model Risk Policy. This reflects: (a) the Model does not itself strike NAV (which would be Tier 1); (b) its output is subject to independent maker-checker sign-off and a further independent Fund Accounting check before it affects any price (Procedure §5); and (c) a modelling error would misallocate cost between two groups of unitholders rather than misstate the fund's total value. Tier 2 status requires annual full revalidation by Model Validation, quarterly performance monitoring reported to the Liquidity Risk Committee, and Model Validation sign-off before any material methodology change (Section 16).

4. Conceptual soundness

4.1 When a fund pays a redeeming investor at net asset value and then sells assets to raise the cash, it incurs a cost — spread, commission, tax, and the price impact of the sale. If that cost is not charged to the redeeming investor, it is borne by every investor who remains, without their consent and without their knowledge, because it never appears in a headline performance number. The anti-dilution mechanism exists to charge that cost to the investor who causes it. Two design principles follow directly from that purpose and are binding on this Model:

(a) The cost must be estimated on a pro-rata basis across the whole portfolio (the "vertical slice"), not on whatever the fund happens to trade. If a fund always sells its most liquid holdings first, the calculated cost

looks small, the redeeming investor is under-charged, and the fund left behind is more concentrated in illiquid assets than the one the remaining investors chose. Pricing the slice removes this ratchet: the estimated cost is independent of which assets are actually sold.

(b) The cost must include market impact, not only the visible spread. A 2020 joint Bank of England / FCA survey of 272 UK funds found that only 13 included market impact in their swing factor calculation. This Model exists to close that gap.

4.2 Theoretical basis for the functional form

4.2.1 The Model decomposes the cost of transacting a position into a spread component and a size-dependent impact component. This decomposition, and the specific functional forms used, are grounded in established market microstructure theory rather than constructed ad hoc.

(a) Half-spread as the floor cost. A fund's net asset value is struck at the mid-market price. A trade can only execute at the touch (best bid or best offer), not at the mid. For an infinitesimally small, uninformed trade, the realised cost relative to the valuation basis converges to exactly half the quoted spread; this is a standard result in the dealer-quoting literature (the effective-spread concept underlying Roll's (1984) implied-spread estimator and the Glosten-Milgrom (1985) framework for spread decomposition into order-processing, inventory and adverse-selection components). It is also the convention used by third-party transaction cost models the Firm references for benchmarking, including Bloomberg's LQA effective-spread formulation, in which effective spread tends to one-half the quoted spread as trade size falls.

(b) Square-root scaling of market impact. Price impact is empirically and theoretically sub-linear in trade size relative to available liquidity: doubling an order does not double its cost. This is consistent with the linear-impact result of Kyle (1985) once expressed cumulatively over an order executed via a sequence of child trades, with the optimal-execution framework of Almgren and Chriss (2001), and with the empirically documented "square-root law" of market impact reported across multiple asset classes and venues (see, for example, Torre (1997, BARRA United States Equity Model documentation) and Tóth et al. (2011)). The Model adopts a square-root exponent (0.5) as the reduced-form approximation of this literature. Section 14 quantifies the sensitivity of the Model's output to this choice, and Section 15 records it as an explicit limitation rather than a firm-specific estimated parameter.

(c) Dealer-quoted markets (fixed income). Government and corporate bonds do not trade on a continuous central limit order book; liquidity is provided by a panel of dealers who quote on request and manage the resulting position through inventory. The relevant theoretical basis is the dealer-inventory literature (Amihud and Mendelson, 1980), under which quoted spread and available size are functions of dealer inventory capacity, issue size, and issue age, rather than of exchange-traded volume. The Model reflects this by sourcing one-day capacity and cost inputs for bonds from dealer-quoted and cross-sectionally modelled data (Section 10, Section 11) rather than from an exchange average-daily-volume calculation, which would not be meaningful for these instruments.

5. The cost model

5.1 For every position, the estimated cost of transacting a given trade size is:

$$\text{cost} = \text{half-spread} + A \times (\text{trade size} \div \text{one-day capacity})^{0.5} + \text{commission [+ transaction tax, buy side only]}$$

Term	Definition and source
Half-spread	Half the effective quoted bid-offer spread, sampled per Section 11.
One-day capacity	For an exchange-traded security (equity, ETF): the Firm's standard 20% participation-rate cap applied to that security's own 60-day median daily traded value. For a bond: the size a panel of dealers would absorb in a day at approximately the quoted spread, a function of issue size, age and sector rather than exchange volume (Section 10, Section 11).
A (impact coefficient)	The market impact, in basis points, incurred when a trade exactly equal to one day's capacity is executed. Estimated per Section 12; reviewed quarterly.
Exponent (fixed at 0.5)	Not separately estimated by security or asset class; see Section 15, Limitation L1.

5.2 Capacity and impact are computed **per security**, never for a sleeve or an asset class. Averaging across a group of holdings understates the cost of the least liquid names in it and overstates the cost of the most liquid, and both errors point the wrong way for the purpose of this calculation.

6. Calculating the fund-level rate: the vertical slice

6.1 For a net fund flow equal to $p\%$ of net asset value, the fund-level anti-dilution rate is the value-weighted sum, across every position in the portfolio, of the cost of transacting $p\%$ of that position:

$$\text{fund rate} = \sum_i (\text{position}_i \div \text{NAV}) \times \text{cost}_i(p \times \text{position}_i)$$

6.2 This is computed for every position, including positions that would not in fact be sold to meet the redemption — the slice is a **pricing basis**, not an execution instruction. The portfolio manager retains discretion over which assets are actually sold; the fund-level rate is unaffected by that choice.

7. Worked example — the Anchorpoint Multi-Asset Income Fund

UK UCITS, SINGLE-PRICED, DAILY DEALING, NAV £500M, 15 POSITIONS

A net redemption of £20m (4% of NAV) requires selling 4% of every line:

Position	Value	4% slice	Slice ÷ capacity	Half-spread	Impact	Cost	Weighted contribution
High yield corporate bond	£30m	£1.20m	0.80×	60.0	49.2	109.2	6.55
EM sovereign (hard ccy)	£20m	£0.80m	0.40×	45.0	25.3	70.3	2.81
UK AIM equity holding	£8m	£0.32m	2.00×	60.0	113.1	175.1	2.80
IG corporate bond — small issue	£30m	£1.20m	0.40×	28.0	13.9	41.9	2.51
European small cap equity	£15m	£0.60m	1.00×	25.0	40.0	67.0	2.01
IG corporate bond — benchmark issue	£45m	£1.80m	0.23×	18.0	6.6	24.6	2.22
High yield bond ETF	£25m	£1.00m	0.20×	20.0	11.2	33.2	1.66
UK mid cap equity	£22m	£0.88m	0.63×	12.0	17.4	31.4	1.38
UK large cap equity	£35m	£1.40m	0.16×	4.0	3.9	9.9	0.70
Developed market equity ETF	£45m	£1.80m	0.06×	3.0	1.0	6.0	0.54
Global mega-cap equity B	£38m	£1.52m	0.05×	2.5	1.6	6.1	0.47
Global mega-cap equity A	£42m	£1.68m	0.04×	2.0	1.2	5.2	0.44
UK gilt 2031	£65m	£2.60m	0.05×	2.0	0.4	2.5	0.32
US Treasury 2030	£35m	£1.40m	0.04×	1.5	0.3	1.8	0.12
Cash and near cash	£45m	£1.80m	—	0.5	—	0.5	0.04
Fund-level swing factor							24.58 bp

Reading it: four positions — 17.7% of the fund — produce 60% of the swing factor. Compare this to the cost of meeting the same £20m redemption entirely from the £45m cash line: **0.5 bp**. The difference, 24.1 bp on a £20m flow, is **£48,160** transferred from remaining to redeeming investors on that single day if the fund is priced the wrong way — and it also leaves the fund with £25m of cash instead of £45m, more concentrated in high yield and emerging market debt than before, with no investment decision having been taken.

8. Scaling with flow size, and buy/sell asymmetry

8.1 Because market impact scales with the square root of trade size, the rate is not linear in flow size. A single fixed factor is therefore wrong at every flow level except the one it was set for. The Firm publishes a continuous schedule, not a stepped one, to avoid a cliff an investor could deal around:

Net flow (% of NAV)	Sell-side rate	Buy-side rate	Note
0.5%	17.9 bp	23.6 bp	Buy-side is higher throughout, entirely because of stamp duty reserve tax on the two UK equity lines; bonds, ETFs, the AIM holding and non-UK equities carry no such asymmetry.
1.0%	19.4 bp	25.1 bp	
4.0%	24.6 bp	30.3 bp	
10.0%	30.6 bp	36.3 bp	

9. Stressed calibration

9.1 The Firm runs the same calculation weekly under a stressed parameter set: half-spreads $\times 3$, impact coefficients $\times 2$, no other change. On the 4% flow example, the sell-side rate moves from **24.6 bp to 61.5 bp** — a multiple of **2.5 $\times$** . This is compared against every fund's disclosed prospectus maximum, because a cap set for normal conditions can silently disable the mechanism at the point unitholders need it most.

9.2 The stress multipliers of $3\times$ (spread) and $2\times$ (impact) are judgemental overlays informed by observed spread and impact widening in the 2020 and 2022 stress episodes for comparable UK multi-asset portfolios. They are not derived from a statistical extreme-value or historical-scenario re-weighting exercise. This is recorded as Limitation L4 in Section 15, together with the Firm's plan to move to a historically-calibrated stress overlay.

THE PROSPECTUS CAP CHECK

Where a fund's disclosed maximum sits below its stressed calibration — for example a 50 bp cap against a 61.5 bp stressed rate — this is escalated to the Committee and to the prospectus review workstream. IOSCO's guidance is explicit that a disclosed range "should not be regarded as a cap or restriction that would prevent anti-dilution [tools] from achieving their objective," and recommends a documented exceedance clause for exceptional, justified circumstances. The Firm's standard prospectus wording includes such a clause; see the Desktop Procedure, Schedule 2.

Every fund's prospectus must reflect the anti-dilution mechanism and policy required by COLL 4.2.5R(18) by the earlier of its next scheduled update or **1 August 2027**, the expiry of the FCA 2026/54 transitional provision. The cap check above is run for every fund now, so that any required change is identified well ahead of that date rather than discovered by it.

10. Asset-class parameterisation

Asset class	One-day capacity source	Impact coefficient (A) source	Calibration tier	Notes
Listed equity	20% of 60-day median traded value	Firm-calibrated from own execution data by market-cap band and region (Section 12.2)	Tier 1 (firm data) where $N \geq 30$ trades; else Tier 2	Median, not mean — a single block trade should not double assumed capacity.
Government bonds	Dealer-quoted absorbable size	Vendor cross-sectional model (LQA), cross-checked quarterly against realised dealer spreads	Tier 2	Spread and impact both widen sharply in stress; government bonds are not immune.
Investment grade credit	Function of issue size and age	Vendor cross-sectional model by rating band and sector, cross-checked against dealer runs	Tier 2	Two bonds of the same rating and sector can have very different capacity if issue sizes differ.
High yield and emerging market debt	As above, materially smaller	Vendor model plus expert judgement overlay where data is thin	Tier 3 — judgement-dependent	Disproportionate contributor to fund-level cost even at modest weights — see Section 7.
ETFs	On-screen average daily volume up to the point a market maker would create or redeem units	Firm-calibrated below that point; underlying-basket cost above it	Tier 1 / Tier 2 (size-dependent)	An ETF is only as liquid as its underlying once the trade exceeds normal secondary-market absorption.
Direct property (multi-asset sleeve)	Not modelled as a daily-tradable capacity	Assumed realisation discount plus professional fees, from comparable transaction evidence	Tier 3 — judgement-dependent	Included in the vertical slice even where not expected to be sold — see Section 4.1.
Money market instruments	Not applicable	Not applicable	Out of scope	Regulated money market funds are outside the anti-dilution mechanism entirely; see the Procedure, Appendix.

11. Data lineage

Input	Source system / field	Refresh	Fallback if unavailable
Quoted bid-offer spread	Primary exchange feed (equities/ETFs); composite dealer contribution (bonds)	Live; 5-day median used for calibration	Prior day's closing quote; if stale >3 days, flagged for manual review
60-day median traded value	Market data vendor; cross-checked against Bloomberg <code>LQA_EXPECTED_DAILY_VOLUME</code> (field PY229)	Daily	30-day window if 60-day history unavailable (recent listing), flagged Tier 2
Implicit cost estimate	Bloomberg LQA <code>LQA_LIQUIDATION_COST</code> (PY251), confidence and participation overrides per Section 12.3	Live, pulled at calibration run-time	Firm's own square-root model using locally estimated A (equities); Illiquid Asset Cost Table (property, thinly-traded credit)
Explicit costs (commission, tax)	Firm execution rate card, maintained by Trading Operations	Reviewed quarterly, on broker fee schedule change	Prior rate card value; escalated if a market's rate is materially stale
Own execution data (for Tier 1 calibration)	Order/Execution Management System, matched to arrival price and post-trade market data	Trailing 12 months, rolling	Not applicable — absence of sufficient data determines the calibration tier, not a fallback value

WORKED ILLUSTRATION — DERIVING ONE-DAY CAPACITY

UK large cap equity line: 60-day median daily traded value £45m × 20% participation cap = **£9m one-day capacity**, matching the value used in Section 7. This derivation is recalculated daily and is available for inspection for any position and any date, per the Desktop Procedure, SOP 3.

12. Parameter estimation methodology

12.1 Half-spread

12.1.1 Half-spread is not estimated; it is observed directly from quoted or contributed market data, per Section 11, and requires no statistical fitting. Its only estimation risk is staleness, controlled by the fallback rule in Section 11.

12.2 Impact coefficient (A) — listed equity and ETFs (Tier 1)

12.2.1 For each security with sufficient own execution history, A is estimated by ordinary least squares regression of realised implementation shortfall against the square root of participation:

$$\text{shortfall}_{bp} = A \times (\text{order size} \div \text{ADV on the day})^{0.5} + \varepsilon$$

12.2.2 Implementation shortfall is measured as the difference between the arrival price (mid-price at order release) and the volume-weighted average execution price, adjusted for the contemporaneous market move using a matched index or peer-basket benchmark, expressed in basis points. The regression is estimated over a trailing 12-month rolling window, re-estimated quarterly.

12.2.3 Sample construction and outlier treatment. Only parent orders (not child slices) are used, to avoid double-counting a single trading decision. Observations are winsorised at the 1st and 99th percentile of shortfall within each market-cap band to limit the influence of data errors, corporate-action-affected prints, and halted-and-

reopened trading. A minimum of 30 qualifying observations in the trailing window is required for a security-specific estimate to be used.

12.2.4 Credibility weighting for thin data. Where a security has fewer than 30 qualifying observations, its estimate of A is a credibility-weighted blend of the security-specific regression coefficient and the market-cap-band and region prior:

$$A_{\text{used}} = z \times A_{\text{security}} + (1-z) \times A_{\text{band}}, \quad z = n \div (n + k)$$

12.2.5 where n is the number of qualifying observations for the security and k is a credibility constant (currently $k = 30$, subject to annual review) chosen so that a security with no trading history relies entirely on the band-level prior, and one with a full 12 months of liquid trading relies predominantly on its own data. This is a standard actuarial (Bühlmann-type) credibility approach applied here to a market-impact parameter rather than a claims-cost parameter, and its purpose is identical: to avoid a security-specific estimate built on too little data from dominating the calibration.

12.2.6 A security with zero qualifying observations (no history of Firm trading, including new listings) is assigned the band-level prior in full, and is flagged in the calibration output as Tier 2 pending accumulation of sufficient execution history.

12.3 Impact coefficient (A) — fixed income (Tier 2 / Tier 3)

12.3.1 No continuous limit order book exists for bonds, so the regression approach in Section 12.2 is not available at security level. A is instead sourced from Bloomberg's fixed income liquidity model (enhanced GSAC methodology), which estimates a latent available-volume distribution per bond from issuer, issue and market characteristics using a cross-sectional model trained across the dealer-quoted universe, retrieved via `LQA_LIQUIDATION_COST` at the security's one-day capacity with confidence level and participation overrides fixed per the Model Parameter Schedule.

12.3.2 This vendor-sourced estimate is cross-checked quarterly against the Firm's own realised dealer execution spreads for completed trades in comparable rating, sector, and issue-size buckets (a bucket-level, not security-level, back-test, given typical trade frequency in any single bond). Where the vendor estimate and the Firm's realised bucket-level average diverge by more than 25%, the security is flagged for Head of Liquidity Risk review and may be assigned a manual override, logged per Section 16.2.

12.3.3 High yield, emerging market and other thinly-traded credit is treated as Tier 3: the vendor model is used as the starting point but is subject to a mandatory expert-judgement review by the Head of Liquidity Risk at every quarterly recalibration, given the greater risk of the vendor model extrapolating beyond its own training data for these instruments (Section 15, Limitation L5).

12.4 Recalibration frequency and governance

12.4.1 All parameters are re-estimated quarterly, in line with the Firm's stress testing cycle, and held between recalibrations in the versioned Model Parameter Schedule. An out-of-cycle recalibration is triggered by a Red backtesting outcome (Section 13.4) and requires the same Liquidity Risk Committee approval as a scheduled recalibration.

13. Backtesting framework

13.1 The Model is backtested quarterly against the Firm's own realised execution costs for every qualifying transaction (parent orders above £250,000 notional) in the trailing quarter. Three tests are applied.

Test	Statistic	Definition	Threshold
T1	Bias	Mean(realised cost – modelled cost) ÷ mean(modelled cost), across all qualifying trades in the quarter	Green: within ±10%. Amber: ±10–20%. Red: outside ±20%.
T2	Coverage / exceedance	Proportion of realised costs exceeding the Model's stated confidence-level prediction (e.g. the 90th percentile stressed estimate), compared to the expected 10% exceedance rate, using a traffic-light exceedance count analogous to standard VaR backtesting	Green: 0–4 exceedances per 40 stressed-regime trades. Amber: 5–9. Red: 10 or more.
T3	Explanatory power	R ² of the Section 12.2 cross-sectional regression, tracked quarter on quarter	Green: R ² ≥ 0.35 and stable. Amber: declining trend over 2 consecutive quarters. Red: R ² < 0.20 or a break in the relationship.

13.2 Green — no action; result reported to the LRC.

13.3 Amber — enhanced monitoring; parameters re-estimated at the next scheduled cycle with the affected population flagged; reported to the LRC with a named owner.

13.4 Red — immediate escalation to the Head of Liquidity Risk and the Chief Risk Officer; an interim conservative override (an additive margin applied to A for the affected asset class or band) is applied within one business day pending root-cause analysis; Model Validation is notified and an out-of-cycle review is initiated; reported to the Board at its next meeting regardless of the outcome of the review.

ILLUSTRATIVE BACKTESTING OUTPUT (FOR FORMAT ONLY — NO LIVE TRADING DATA HAS YET ACCUMULATED UNDER THIS VERSION OF THE MODEL)

Asset class	N (trades)	T1 Bias	T1 Status	T3 R ²	T3 Status
UK / global large cap equity	146	+4.1%	GREEN	0.47	GREEN
UK mid / small cap equity	38	–12.6%	AMBER	0.31	GREEN
Investment grade credit	22	+7.8%	GREEN	n/a — bucket test	—
High yield / EM credit	9	–18.4%	AMBER	n/a — bucket test	—

This table is a template for the report the Head of Liquidity Risk produces each quarter once a full quarter of post-implementation trading data exists; it is not a claim about the Firm's actual historical performance. The mid/small cap equity and high yield/EM bias readings shown here are the pattern most commonly observed by Model Validation in comparable models at first backtest — systematic under-recovery in the least liquid names — and this template is deliberately populated to show the Committee what an Amber outcome looks like and what it requires, rather than to be read as a clean result.

14. Sensitivity analysis

14.1 The following shows the effect on the fund-level swing factor (Section 7 worked example, 4% net redemption, all other parameters held at base case) of a ±20% change in each key parameter, and of the fixed impact exponent used in place of the literature midpoint. All figures independently recomputed from the Section 7 dataset.

Parameter change	Swing factor	Δ vs base (24.58bp)	Interpretation
Base case	24.58 bp	—	—
Half-spread inputs +20%	27.22 bp	+2.64 bp	The Model is materially more sensitive to spread data quality than to the impact coefficient — stale or wide quotes move the output more than a plausible impact misestimate.
Half-spread inputs –20%	21.94 bp	–2.64 bp	Symmetric, as expected for a linear term.
Impact coefficient (A) +20%	26.66 bp	+2.08 bp	A 20% error in impact calibration moves the fund rate by roughly 8%.
Impact coefficient (A) –20%	22.50 bp	–2.08 bp	Symmetric.
Participation cap halved (10% not 20%, equity/ETF only)	26.62 bp	+2.04 bp	A more conservative capacity assumption raises the rate — intuitive, since less daily capacity means each unit of flow is a larger share of it.
Participation cap doubled (40% not 20%, equity/ETF only)	23.14 bp	–1.44 bp	Asymmetric response versus the halving case, because capacity enters inside a square root.
Impact exponent 0.4 (vs. base 0.5)	25.13 bp	+0.55 bp	A lower exponent raises cost for the smallest slices (sub-unity ratios) and lowers it for the largest — net effect here is small and positive.
Impact exponent 0.6 (vs. base 0.5)	24.13 bp	–0.45 bp	The Model's output is not highly sensitive to the exponent choice at this fund's actual position sizes, which supports (but does not eliminate the need to disclose) the fixed-exponent simplification at Limitation L1.

14.2 The dominant sensitivity is to the half-spread input, which is an observed market data item rather than an estimated parameter; this places a premium on the data quality and staleness controls in Section 11, more so than on the precision of the impact coefficient regression in Section 12.

15. Model limitations

15.1 The following limitations are known and accepted by the model owner and the Liquidity Risk Committee as at the effective date of this document. Each is monitored; none is considered to invalidate the Model for its intended use.

Ref	Limitation	Nature and materiality	Mitigation / monitoring
L1	Fixed impact exponent (0.5)	Not re-estimated per security or asset class; the literature range is approximately 0.4–0.6. Section 14 shows the resulting sensitivity is modest at this fund's position sizes but may not be at larger sizes or in more concentrated funds.	Monitored via T3 (Section 13); a firm-specific exponent estimation project is on the Model Validation-agreed roadmap for the next full revalidation cycle.
L2	Fixed income capacity is judgement-informed, not order-book-derived	Dealer-quoted absorbable size is inherently less precise than an exchange-observed capacity and may be procyclically overstated by dealers in calm markets.	Quarterly cross-check against realised bucket-level execution (Section 12.3.2); Tier 3 expert-judgement overlay for high yield/EM.
L3	No cross-fund / aggregation effect	The Model prices each fund's flow independently. Where the Firm's other funds hold the same or correlated positions and are also transacting, true available capacity is lower than the Model assumes for any single fund.	Addressed operationally, not within this Model, by the cross-fund aggregation overlay in the Desktop Procedure (SOP 5); flagged here so Model Validation does not expect this Model alone to capture it.
L4	Stress multipliers are judgemental overlays	The 3× spread / 2× impact stress scalars (Section 9) are informed by past episodes but are not derived from a formal extreme-value or historical-scenario re-weighting methodology.	Reviewed annually against realised stress-period data as it accumulates; a statistically-derived stress overlay is planned as a methodology enhancement.
L5	Vendor model extrapolation for thin-data instruments	The fixed income cross-sectional model will return an estimate for a bond with little or no trading history; this is a strength for coverage but a risk if relied upon uncritically.	Tier 3 classification and mandatory expert-judgement review (Section 12.3.3); flagged in all calibration outputs.
L6	Independence assumption between spread and volume shocks	The stress overlay applies spread widening and volume contraction as independent multiplicative scalars; in practice they are correlated, and the combined effect in a genuine liquidity crisis may exceed the product of the two independent scalars.	Conservative overlay sizing partially compensates; flagged as a direction for future enhancement alongside L4.
L7	Credibility constant ($k = 30$) is not itself statistically fitted	The credibility weighting in Section 12.2.4 uses a round-number constant chosen for interpretability rather than fitted to minimise out-of-sample estimation error.	Annual review; a formally fitted credibility constant is a candidate methodology enhancement.

16. Ongoing monitoring and change control

16.1 Performance is monitored continuously through the backtesting framework (Section 13) and reported to the LRC quarterly and to the Board semi-annually as part of the wider liquidity reporting cycle.

16.2 Overrides. Any replacement of a Model output by expert judgement (Sections 12.3.2, 12.3.3) is logged in the Override Log with the security, the Model output, the substituted value, the reason, and the approving individual, and is reviewed quarterly by the LRC for systematic, one-directional patterns that would indicate a calibration weakness rather than isolated judgement calls.

16.3 Change control. A change to a live parameter within the existing methodology (e.g. a quarterly recalibration of A) requires Head of Liquidity Risk sign-off and LRC notification. A change to the methodology itself (e.g. the functional form, the credibility constant, the stress multipliers, or the data source for any input) requires Model Validation review and sign-off before it may be adopted, and is treated as a new model version for the purposes of the Model Risk Policy.

16.4 Vendor dependency. A material change to the Bloomberg LQA methodology is a change-control event under this section: it requires re-validation, quantification of its effect on rates, buckets and thresholds across the fund range, and LRC approval before adoption, notwithstanding that the change originates outside the Firm.

17. Model risk rating and validation record

INDEPENDENT MODEL VALIDATION OPINION

Validation status	[Pending first submission — to be completed by Model Validation]
Conceptual soundness	[Validator assessment]
Implementation testing	[Validator assessment]
Ongoing monitoring adequacy	[Validator assessment]
Overall model risk rating	[High / Medium / Low — to be assigned by Model Validation]
Conditions of use / restrictions	[Any use restrictions Model Validation imposes pending remediation of a limitation in Section 15]

This document is submitted to Model Validation in this form. The Liquidity Risk Committee will not approve the Model for live use, and will not rely on its output for the pricing of any fund, until Model Validation has completed the assessment above and any conditions of use have been satisfied or accepted by the Committee. The template above is intentionally left for completion by the independent validator and must not be pre-populated by the model owner.

Model owner

Independent validation

Head of Liquidity Risk

Model Validation

Approved by the Liquidity Risk Committee under the authority of the Fund Liquidity Risk Policy, Section 6.3, following Model Validation sign-off. All figures in the worked example, sensitivity analysis and illustrative backtesting output are internally consistent and independently verified against the stated formulae; the illustrative backtesting output in Section 13 is a reporting template, not a record of the Firm's actual historical performance.